

One Involution: The Recognition Cost and the Golden Ratio from a Single Symmetry

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Abstract

The Recognition Science program treats two quantities as fundamental: the recognition cost $J(x) = \frac{1}{2}(x + x^{-1}) - 1$, and the golden ratio φ that sets the scale of its self-similar hierarchy. These look like two independent inputs. They are not. Both are determined by a single object, the reciprocal involution $\iota(x) = x^{-1}$ on the positive reals. The cost is exactly the ι -symmetric cost, and the fixed point of ι is precisely the cost's unique zero, the unit. The golden ratio is exactly the unique fixed point greater than one of the affine shift $1 + \iota$, the map obtained by perturbing ι by its own fixed point. So the unit and the scale, the two constants the framework is built from, are the two fixed points of one involution family $\{\iota, 1 + \iota\}$. We state this precisely, prove it, and are explicit about what it does and does not claim. Every assertion is machine-checked in Lean 4 against Mathlib, with no `sorry` and no Recognition-Science-specific axiom; the module is `IndisputableMonolith.Foundation.UniversalForcing.ReciprocalGenerator`.

1 The claim, and why it is not obvious

A reader who has seen the cost J and the golden ratio φ derived separately in the Recognition Science literature could reasonably take them to be two distinct primitives: one a function pinned by a functional equation, the other a number pinned by a self-similarity constraint. The cost-uniqueness theorem fixes J from reciprocal symmetry, normalization, a composition law, and calibration. The hierarchy theorem fixes φ from self-similar scaling and additive closure. Two theorems, two hypothesis packages, two constants. Nothing in that account forces the two to share an origin.

This note shows they do, in a sharp and checkable sense. The shared origin is the reciprocal involution

$$\iota: \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}, \quad \iota(x) = x^{-1}.$$

This is the duality that exchanges a quantity with its reciprocal: the dual-recognition symmetry. It is the only structural ingredient we will use. From it, both the cost and the scale fall out as fixed-point data, and the two constants turn out to be the two fixed points of a single one-parameter family of maps built from ι .

The argument is short because the content is structural rather than computational. What makes it worth stating is not difficulty but identification: it removes the appearance that J and φ are independent posits.

2 The involution

Definition 2.1 (Reciprocal involution). $\iota(x) = x^{-1}$ on the positive reals.

Proposition 2.2. ι is an involution on $\mathbb{R}_{>0}$: $\iota(\iota(x)) = x$ for $x > 0$. Its unique fixed point is the unit: for $x > 0$, $\iota(x) = x$ if and only if $x = 1$.

Proof. $\iota(\iota(x)) = (x^{-1})^{-1} = x$. If $x^{-1} = x$ then $x^2 = 1$, and the only positive root is $x = 1$; conversely $1^{-1} = 1$. $\square$

The fixed point of ι is the unit. Hold onto this. It is the hinge of the whole note: the unit is not chosen separately from ι , it is read off ι as the one place the reciprocal symmetry leaves invariant.

3 Cost side: J is the ι -symmetric cost, and its zero is ι 's fixed point

The recognition cost is built directly on ι . Writing it as

$$J(x) = \frac{1}{2}(x + \iota(x)) - 1$$

makes the dependence explicit: J is the symmetric average of a quantity and its reciprocal, recentered so the unit costs nothing.

Proposition 3.1 (Cost symmetry). For $x > 0$, $J(\iota(x)) = J(x)$.

Proof. $J(x^{-1}) = \frac{1}{2}(x^{-1} + x) - 1 = J(x)$. Equivalently, written as a square, $J(x) = (x - 1)^2/(2x)$ is manifestly invariant under $x \mapsto x^{-1}$ after clearing denominators. $\square$

So J is invariant under the reciprocal involution: it is constant on ι -orbits $\{x, x^{-1}\}$. This is the first hypothesis of the cost-uniqueness theorem, and it is the structural reason J has the shape it has. The sharper fact is that ι 's fixed point and J 's zero are the same point.

Theorem 3.2 (The involution's fixed point is the cost's zero). For $x > 0$,

$$\iota(x) = x \iff J(x) = 0,$$

both holding exactly at $x = 1$.

Proof. By Proposition 2.2, $\iota(x) = x$ iff $x = 1$. By the squared form $J(x) = (x - 1)^2/(2x)$, for $x > 0$ the cost vanishes iff $x = 1$. The two conditions therefore coincide. $\square$

This is more than “ J is ι -symmetric.” The symmetry axis of ι , the locus it fixes, is identical to the null set of J , the locus of zero recognition cost. The unit is simultaneously the only self-reciprocal positive number and the only costless one. The cost does not introduce a new distinguished point; it inherits ι 's.

4 Scale side: φ is the unique fixed point of $1 + \iota$

The golden ratio enters through the same involution, shifted. Consider the map

$$g(x) = 1 + \iota(x) = 1 + x^{-1}.$$

The shift is by 1, which is exactly ι 's own fixed point (Proposition 2.2). So g is the involution perturbed by the unit it already singles out. Its fixed-point equation is the self-similarity constraint.

Theorem 4.1 (φ is the unique fixed point of $1 + \iota$ above one). For $x > 1$,

$$g(x) = x \iff x = \varphi,$$

where $\varphi = (1 + \sqrt{5})/2$ is the golden ratio.

Proof. For $x > 0$, $g(x) = x$ reads $1 + x^{-1} = x$. Multiplying by x gives $x + 1 = x^2$, that is $x^2 = x + 1$. The positive roots of $x^2 = x + 1$ are φ and $(1 - \sqrt{5})/2 < 0$, so among $x > 1$ the unique solution is φ . Conversely φ satisfies $\varphi^2 = \varphi + 1$, hence $\varphi = 1 + \varphi^{-1} = g(\varphi)$. $\square$

The self-similarity that fixes the scale is not a second principle layered on top of the cost. It is the fixed-point condition for the same involution, displaced by the unit. Where ι fixes the unit, $1 + \iota$ fixes φ .

5 One involution, two constants

The two constants now sit in a single picture.

Theorem 5.1 (The shared generator). *The reciprocal involution ι determines both fundamental quantities through fixed points:*

- (i) *the fixed point of ι is the unit 1, which is exactly the unique zero of the recognition cost J ;*
- (ii) *the fixed point of the shift $1 + \iota$ above one is the golden ratio φ , the scale.*

Proof. Immediate from Theorems 3.2 and 4.1. $\square$

The unit and the scale, the two numbers the framework is built from, are the two fixed points of one involution family $\{\iota, 1 + \iota\}$. The family is not arbitrary either: its two members are ι and ι shifted by ι 's own fixed point, so the whole structure is generated by ι alone with no second input. The cost is the ι -symmetric average; φ is the fixed point of the minimal shift of ι by the unit it fixes.

Machine verification. Every statement above is proved in Lean 4 against Mathlib in the module `IndisputableMonolith.Foundation.UniversalForcing.ReciprocalGenerator`. Proposition 2.2 is `recip_involutive` and `recip_fixed_iff`; Proposition 3.1 is `jcost_recip_symmetric`; Theorem 3.2 is `recip_fixed_iff_cost_zero`; Theorem 4.1 is `recipShift_fixed_iff` with witness `phi_is_recipShift_fixed`; Theorem 5.1 is `recip_pins_unit_and_scale`. The bundle is recorded as `ReciprocalGeneratorCert / reciprocalGeneratorCert_holds`. A `#print axioms` on these declarations reports no `sorry` and no Recognition-Science-specific axiom; the cost-symmetry and golden-ratio facts the proofs invoke (`Cost.Jcost_symm`, `Cost.Jcost_eq_zero_iff`, `PhiSupport.phi_unique_pos_root`) are themselves elementary real algebra.

6 What this does, and does not, claim

The claim is an identification of origin, and it should be read at exactly that strength, no more.

It does claim that J and φ are not independent posits. Both are fixed by the single involution ι , the cost by symmetry and the scale by the shifted fixed point, with the unit serving as the literal hinge between the two: it is ι 's fixed point, it is J 's zero, and it is the amount by which ι is shifted to produce the map that fixes φ .

It does not claim that the entire cost-uniqueness theorem is subsumed. Pinning J to its exact functional form needs more than ι -symmetry: it needs normalization, the composition law, calibration, and continuity. ι -symmetry is one hypothesis among those, the one that fixes the symmetry axis; the others fix the profile along it. This note isolates the shared hypothesis, not the full derivation.

It does not claim that the full φ -forcing chain is subsumed either. Deriving φ as the scale of a physical hierarchy needs the additive-closure and self-similarity content of that hierarchy; what this note shows is that the *equation* those principles produce, $x = 1 + x^{-1}$, is the fixed-point

equation of $1 + \iota$, so the constant they land on is ι -generated. The physics that selects this equation is upstream and is not relitigated here.

And it does not claim that physics is reduced to a reciprocal symmetry. The honest reading is narrower and, for that reason, defensible: two quantities the framework had treated as separate fundamentals are two fixed points of one involution. That is a structural unification of inputs, not a theory of everything. It is the kind of result that makes a framework smaller, which is the direction a foundational program should want to move.